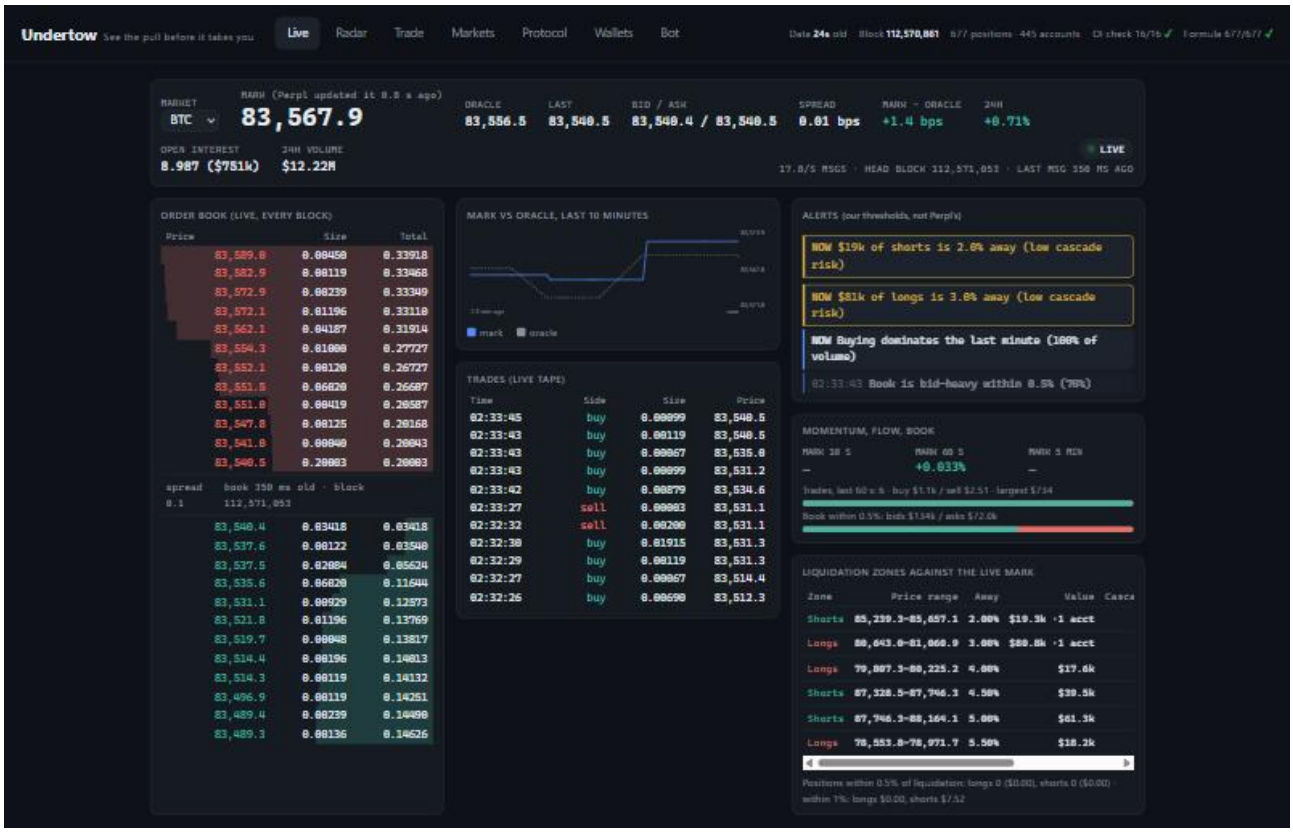
# Undertow - the live terminal

The first tab of the dashboard. Built and tested on 12 October 2026 against Perpl mainnet.

## What it is

A screen that updates with every Monad block (about every 0.4 seconds) using **Perpl's own public WebSocket feed**: oracle, mark and last price, the order book, and every trade. It is not our estimate of those things; it is what Perpl broadcasts. On top of that feed it computes risk signals against the liquidation-zone data, re-measuring every zone against the **live** mark price on each update.




Live at build time: 18.0 messages per second, head block 112,571,585, last message 184 ms old.

## What is on the screen

| Panel | Source and update speed |
|---|---|
| Ticker bar: mark, oracle, last, bid/ask, spread, mark minus oracle, 24h change, open interest, volume | Perpl feed, pushed on change. Perpl itself updates the mark price every few seconds; the screen shows how many seconds ago |
| Order book ladder, 12 levels a side with depth bars | Perpl feed: a snapshot, then every changed level each block |
| Trade tape (big trades highlighted) | Perpl feed, each trade as it happens |
| Mark versus oracle chart, last 10 minutes | Built from the feed as it arrives |
| Momentum (10 s, 60 s, 5 min), buy/sell flow over 60 s, book imbalance within 0.5% | Calculated from the feed on every poll |
| Liquidation zones, distance to the live mark, cascade estimate using the live book, positions within 0.5% of liquidation | Positions from the snapshot (up to about a minute old), measured against the live mark and live book |
| Alerts (zone close, fast move, mark far from oracle, one-sided flow, lopsided book, positions near liquidation) | Rule-based, thresholds listed in the code; an alert appears when a condition becomes true and stays logged |

# How accurate is it?

Each line is a real test run just now through the dashboard's checks (Protocol tab, 'Run checks').

| Check | Result | Detail |
|---|---|---|
| ticker: mark price | PASS | ours 83,549.80 vs Perpl 83,549.80 (0.000% apart) |
| ticker: open interest | PASS | ours 8.95287 vs Perpl 8.95287 (0.000% apart) |
| order book: top 10 levels per side | PASS | 20/20 price levels present in both, 20/20 identical in size and order count (both read at block 112,571,312) |
| volume: 20 whole minutes, worst difference 0.00% | PASS | 10-11 18:20Z ours $906 vs Perpl $906 (0.00%, 11 fills vs 10 trades); 10-11 18:21Z ours $5,452 vs Perpl $5,452 (0.00%, 41 fills vs 11 trades); 10-11 18:22Z ours $7,237 vs Perpl $7,237 (0.00%, 29 fills vs 11 trades); 10-11 18:23Z ours $13,582 vs Perpl $13,582 (0 |
| live feed: connected and receiving | PASS | 17.2 messages/s, head block 112,571,491, last message 165 ms ago |
| live feed: mark, oracle, last vs REST ticker | PASS | mark 0.00 bps apart, oracle 0.00 bps apart, last 0.00 bps apart (two reads a moment apart) |
| live feed: order book vs the chain, same block | PASS | 20/20 top-10 levels identical in price, size and order count to the contract state read at block 112,571,482 (WS message block 112,571,483, offset -1); tried offsets {0: 16, -1: 20} |

Run at 2026-10-11 21:06:05 UTC.

In plain words: the prices on the Live tab equal Perpl's own ticker, and the order book kept from the feed equals the book stored in Perpl's contract on the chain, level for level (price, size and number of orders). One detail: Perpl labels a book with the block it was taken at the *start* of, so it equals the contract state read at the previous block. In this run the match was 20 of 20 at that earlier block and 16 of 20 at the labelled block, the same convention found for Perpl's REST book. Those are the parts that must be exact, and in this test they are.

## What 'signals' means here

The **data** is accurate. The **signals** are rules applied to that data, such as 'a $300k zone is 1% from the live mark'. They are risk flags, not predictions of where the price will go, and not trading advice. I did not backtest any of them, and the thresholds are my choices.

## Honest limits

- **Zones are not real-time.** Perpl's feed carries prices, book and trades, not everyone's positions. Positions come from our snapshot (refreshed about every 45 seconds). A position opened or closed in the last minute is missing or stale, so a zone alert can be slightly off. Making positions real-time means rebuilding Perpl's state from every contract event, which is a larger project and is not built.
- **The mark price moves in steps.** Perpl updates it every few seconds, and liquidations use it. The screen shows its age so this is visible rather than looking like lag.
- **Only the 7 busiest markets stream a live book.** Perpl allows 16 subscriptions per connection; each market's book and trades take two.
- **Display delay is at most half a second:** the page asks the server for the latest state twice a second.
- **It needs a working internet path to Perpl's WebSocket.** If the computer sleeps or the connection drops it shows 'FEED STALLED / RECONNECTING' and reconnects by itself, but I tested it for minutes, not hours, so I have not proven long-run stability.
- **The browser tab must be visible.** The page pauses polling when its tab is hidden, to save work.
- The cascade estimate shown for each zone is the same labelled worst-case model as on the Radar tab, now run against the live book. It has not been checked against a real historical liquidation.

## How to use it

Run `start_dashboard.bat` (or `python -m undertow.server --refresh`). The Live tab opens first. It needs one extra package: `pip install websockets`. Code: undertow/live.py and undertow/signals.py, with 14 new tests.